Fifteen points determine at least nine ordinary lines

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Abstract

An *ordinary line* of a finite point set in the real plane is a line containing exactly two of the points. Let $m(n)$ be the minimum number of ordinary lines determined by n points, not all collinear. The values $m(n)$ are known for $n \le 14$ and for the even $n \le 24$, where the Csima–Sawyer bound $\lceil 6n/13 \rceil$ is attained by Böröczky's examples, while for $n = 15$ only $7 \le m(15) \le 9$ has been known (the lower bound is the Kelly–Moser bound of 1958); the entry $a(15)$ of OEIS sequence A003034 is the first unknown one. We prove that $m(15) = 9$, i.e. every set of 15 points, not all on a line, determines at least 9 ordinary lines; Böröczky's example (a regular octagon together with seven of the eight points at infinity of its sides and diagonals) shows that 9 is attained. Thus the Green–Tao formula (proved for all sufficiently large n) holds for all $6 \le n \le 16$ except $n = 13$, and 15 points do not behave like McKee's sporadic 13-point set. The proof is a computer-assisted classification. Melchior's inequality and a counting argument reduce the question to the exact geometric realizability of the partial linear spaces on 15 points with one of seven prescribed line-size profiles (one with 7 and six with 8 ordinary lines). We enumerate these up to isomorphism (23,174,731 isomorphism classes in all) and show that none is realizable in the real projective plane with exactly the prescribed collinearities, using the Pappus and Desargues theorems as combinatorial filters and exact Gröbner-basis computations for the remaining 27,441 classes; all but one of the systems are rejected by arguments valid over every field of characteristic 0. The single remaining system, with three 4-point lines, one 5-point line and 23 three-point lines, has an exact realization over $\mathbb{Q}(\sqrt{-3})$—so 15 points of the complex projective plane can have 8 ordinary lines with a "real-looking" line profile—but it contains the Möbius–Kantor configuration and hence has no real realization. Every algebraic rejection is confirmed by an independent SAT-based test for the existence of a rank-3 oriented matroid, and the same pipeline recovers the known extremal configurations for 12, 13 and 14 points (and shows that they are the only ones), which serves as a positive control.

1 Introduction

Sylvester's problem (1893), solved by Gallai (1944) and by Melchior (1941), asserts that every finite set of points in the real plane, not all on a line, determines at least one *ordinary line*, that is, a line passing through exactly two of the points. Dirac [5] and Motzkin [12] asked for the minimum number $m(n)$ of ordinary lines determined by n non-collinear points, and Dirac conjectured $m(n) \ge n/2$; this fails for $n = 7$ and $n = 13$, and the Dirac–Motzkin conjecture in its modern form asserts $m(n) \ge n/2$ for all other n. Kelly and Moser [9] proved $m(n) \ge 3n/7$, which is sharp for $n = 7$; Csima and Sawyer [3] improved this to $m(n) \ge 6n/13$ for $n \ne 7$, which is sharp for $n = 13$ (McKee's configuration [2]); and Green and Tao [6] proved the Dirac–Motzkin

conjecture for all $n \geq n_0$, together with the exact formula

$$m(n) = \begin{cases} n/2 & n \text{ even}, \\ 3\lfloor n/4 \rfloor & n \text{ odd}, \end{cases} \qquad n \geq n_0, \tag{1}$$

where n_0 is an unspecified, very large constant. The upper bounds in (1) come from Böröczky's examples: the vertices of a regular m-gon together with the m points at infinity in the directions of its sides and diagonals form a set X_{2m} with exactly m ordinary lines; adding the centre gives $4k+1$ points with $3k$ ordinary lines when $m = 2k$, and deleting a suitable point at infinity (one in the direction of a diagonal) gives $4k-1$ points with $3k-3$ ordinary lines.

For small n the exact values are

n	3	4	5	6	7	8	9	10	11	12	13	14
$m(n)$	3	3	4	3	3	4	6	5	6	6	6	7

(OEIS A003034 [13]; see [2, 3] and the references there: the lower bounds come from the Kelly–Moser theorem, from case analysis for $n = 9, 11$, and from the Csima–Sawyer theorem for $n = 14$; the upper bounds from explicit configurations). For $n \geq 15$ the situation is as follows. The Kelly–Moser bound gives $m(15) \geq \lceil 45/7 \rceil = 7$ (the Csima–Sawyer bound gives the same), and Böröczky's example X_{16} minus a point at infinity gives $m(15) \leq 9$. For even $n \leq 24$ the Csima–Sawyer bound is attained by X_n, so $m(16) = 8$, $m(18) = 9$, $m(20) = 10$, $m(22) = 11$, $m(24) = 12$, but none of the values $m(15)$, $m(17)$, $m(19)$ has been determined, and $m(15) \in \{7, 8, 9\}$ is the first open case. Formula (1) predicts $m(15) = 9$. We prove that this is correct.

Theorem 1.1. *Every set of* 15 *points in the real plane, not all on a line, determines at least* 9 *ordinary lines. Hence* $m(15) = 9$.

The classification behind the proof yields a little more.

Theorem 1.2. *Exactly one of the 23,174,731 partial linear spaces on* 15 *points with the seven profiles of Table 1 has an exact realization over some field of characteristic* 0. *It has profile* $t_2 = 8$, $t_3 = 23$, $t_4 = 3$, $t_5 = 1$, *its realization space (normalised on a projective frame) consists of two points defined over* $\mathbb{Q}(\sqrt{-3})$ *and conjugate under complex conjugation, and it contains four copies of the Möbius–Kantor configuration* (8_3). *In particular there is a set of* 15 *points in* $\mathbb{CP}^2$ *with exactly* 8 *ordinary lines,* 23 *three-point lines, three* 4*-point lines and one* 5*-point line, but no such set exists in* $\mathbb{RP}^2$.

Theorem 1.1 says that $n = 15$ is not a sporadic exception to (1), in contrast to $n = 13$ (formula (1) now holds for all $6 \leq n \leq 16$ except $n = 13$, and for the even $n \leq 24$), and it settles the first unknown term of OEIS A003034. Since only the Kelly–Moser bound is used in the reduction, the proof is independent of the Csima–Sawyer theorem. The result also fits the pattern that the sets with few ordinary lines are governed by cubic curves: the extremal example for $n = 15$ lies on the union of a conic and a line, as all Böröczky examples do.

The proof is a computer-assisted classification in the spirit of Crowe and McKee's case analysis, of Bokowski and Pokora's classification of pseudoline arrangements with many triple points [1], and of the recent determination of the orchard-planting number $t_3(15) = 31$ [14]. Its logical structure is as follows.

1. Writing t_k for the number of lines containing exactly k of the points, a configuration of 15 points with $t_2 \le 8$ satisfies $t_2 \in \{7, 8\}$, and the counting identity $\sum_k \binom{k}{2} t_k = 105$ together with Melchior's inequality $t_2 \ge 3 + \sum_{k \ge 4}(k-3)t_k$ leaves exactly seven possible *profiles* $(t_3, t_4, \dots)$, listed in Table 1. The lines with at least three points form a *partial linear space* with these block sizes, and the ordinary lines are exactly the pairs of points not covered by a block (Section 2).

2. We enumerate, up to isomorphism, all partial linear spaces on 15 points with each of the seven profiles: there are 23,174,731 of them (Section 3).

3. For each of them we decide whether it has an *exact realization*, i.e. 15 distinct points of $\mathbb{RP}^2$ whose collinear triples are precisely the triples inside blocks. The Pappus and Desargues theorems, applied to all their instances inside the partial linear space, reject all but 27,441 of the systems; an exact algebraic test (a projective frame, a construction sequence, a Gröbner basis of the incidence ideal and saturation by the non-degeneracy conditions) rejects all of the rest except the one system of Theorem 1.2, whose realizations are computed explicitly and are not real (Sections 4–6).

Section 7 describes the consistency checks: each enumeration stage is checked against an independent implementation and against orbit–stabiliser counts of labelled objects, the closure filters are checked witness by witness and against an unfiltered algebraic run, every algebraic rejection is confirmed both by an independently written Gröbner-basis script and by a SAT-based test for the existence of a rank-3 oriented matroid (a pseudoline arrangement) with the prescribed collinearities, and the pipeline applied to $n = 12, 13, 14$ points with $m(n)$ ordinary lines finds exactly the known extremal configurations, with exact coordinates that we verify independently. All programs and data accompany the paper (Section 9).

Remark 1.3. Four of the seven profiles (the one with $t_2 = 7$ and three with $t_2 = 8$) attain equality in Melchior's inequality; a configuration with such a profile would be dual to a *simplicial* arrangement of 15 lines. Grünbaum's catalogue of simplicial arrangements [7], whose completeness up to 27 lines was established by Cuntz [4] (up to four additional arrangements with 22–25 lines found there), lists six arrangements of 15 lines (the near-pencil and $A(15,1)$–$A(15,5)$), with 14, 15, 13, 12, 12 and 9 double points respectively, so these four profiles could also be excluded by appealing to that classification; our enumeration treats them directly and therefore confirms independently that no simplicial arrangement of 15 lines has fewer than 9 double points. The other three profiles with $t_2 = 8$ are "nearly simplicial" (the dual arrangement has one, two or three non-triangular faces, counted with multiplicity), and no classification is available for them. Note also that $A(15,5)$, with $t_2 = 9$, $t_3 = 22$, $t_5 = 3$, is dual to a second 15-point configuration with 9 ordinary lines, not projectively equivalent to Böröczky's (whose profile is $t_2 = 9$, $t_3 = 25$, $t_7 = 1$).

2 Reduction to partial linear spaces

Throughout, a *configuration* is a finite set P of points of the real projective plane $\mathbb{RP}^2$, not all on a line, and $t_k = t_k(P)$ is the number of lines containing exactly k points of P. Since some line misses all points of a finite set, every configuration is projectively equivalent to one in the affine plane with the same numbers t_k, so the affine and projective versions of Theorem 1.1 are equivalent.

Lemma 2.1. *For a configuration of n points,*

$$\sum_{k\ge 2}\binom{k}{2}\,t_k=\binom{n}{2}, \tag{2}$$

$$t_2\ge 3+\sum_{k\ge 4}(k-3)\,t_k \quad \textit{(Melchior [11])}, \tag{3}$$

$$t_2\ge 3n/7 \quad \textit{(Kelly–Moser [9])}. \tag{4}$$

Identity (2) counts pairs of points; (3) is Euler's formula applied to the dual line arrangement, with equality if and only if all faces of the dual arrangement are triangles. (The stronger Csima–Sawyer bound $t_2\ge 6n/13$ is not needed for $n=15$.)

Lemma 2.2. *Let P be a configuration of 15 points with $t_2\le 8$. Then $t_2\in\{7,8\}$, and $(t_2;t_3,t_4,\dots)$ is one of the seven profiles of Table 1.*

Proof. By (4), $t_2\ge 45/7$, so $t_2\ge 7$. Put $w=\sum_{k\ge 4}(k-3)t_k$; by (3), $w\le t_2-3\le 5$, so $t_k=0$ for $k\ge 9$, and (2) reads $3t_3+6t_4+10t_5+15t_6+21t_7+28t_8=105-t_2$. For $t_2=7$ the right-hand side is 98, and the only choice of $(t_4,\dots,t_8)$ with $w\le 4$ making $98-6t_4-10t_5-\dots$ a non-negative multiple of 3 is $t_5=2$, giving $t_3=26$. For $t_2=8$ the right-hand side is 97 and $w\le 5$; going through the $(t_4,\dots,t_8)$ with $w\le 5$ (there are 19 of them) one finds exactly the six listed profiles. The enumeration of profiles is also carried out mechanically by `profiles.py`. □

t_2	profile $(t_3,t_4,\dots)$	Melchior slack	placements	(P,L) classes
7	$t_3=26,\ t_5=2$	0	2	247
8	$t_3=23,\ t_8=1$	0	1	5
8	$t_3=29,\ t_5=1$	3	1	60
8	$t_3=24,\ t_5=1,\ t_6=1$	0	2	210
8	$t_3=27,\ t_4=1,\ t_5=1$	2	2	814
8	$t_3=25,\ t_4=2,\ t_5=1$	1	7	13,280
8	$t_3=23,\ t_4=3,\ t_5=1$	0	28	237,615

Table 1: The seven profiles of a 15-point configuration with at most 8 ordinary lines (Lemma 2.2); the Melchior slack is $t_2-3-\sum_{k\ge 4}(k-3)t_k$. The last two columns give the numbers of isomorphism classes of placements of the big blocks (Stage A) and of pairs (big blocks, leave) (Stage B) of Section 3.

The profile with an 8-point line is impossible for an elementary reason.

Lemma 2.3. *If a configuration of n points has a line ℓ containing k of the points, and $r=n-k\ge 1$, then $t_2\ge r\,(k-r+1)$.*

Proof. Let $p\notin\ell$. The k lines joining p to the points of ℓ are distinct, and each of the $r-1$ other points off ℓ lies on at most one of them, so at least $k-(r-1)$ of these lines contain no point of P other than p and their point on ℓ: they are ordinary. An ordinary line of this kind contains exactly one point off ℓ, so the ordinary lines obtained from different p are different. □

For $n=15$, $k=8$, $r=7$ the lemma gives $t_2\ge 14$, so the profile $(t_3,t_8)=(23,1)$ does not occur. (The enumeration of Section 3 confirms this: no partial linear space with this profile exists at all.)

Definition 2.4. A *partial linear space* (PLS) on a finite set X is a set $\mathcal{B}$ of subsets of X of size at least 3, called *blocks*, such that two distinct blocks share at most one point. Its *profile* is the sequence $(t_3, t_4, \dots)$ of the numbers of blocks of each size, its *leave* L is the graph on X whose edges are the pairs of points not contained in a block, and its *collinear triples* are the 3-subsets of blocks. An *exact realization* of $(X, \mathcal{B})$ over a field K is an injective map $X \to K\mathbb{P}^2$ such that a triple of points of X is collinear if and only if it is a collinear triple of $\mathcal{B}$.

If P is a configuration, the lines with at least three points form a PLS on P (the *collinearity structure* of P) whose profile is $(t_3(P), t_4(P), \dots)$ and whose leave has exactly $t_2(P)$ edges; the identity map is an exact realization of it. Conversely, let ρ be an exact realization of a PLS $\mathcal{B}$ with profile $(t_3, t_4, \dots)$. A line containing at least three points of $\rho(X)$ contains a collinear triple, which lies in a block B; every further point y of X on the line forms a collinear triple with two points of B, so y lies in a block sharing two points with B, i.e. in B. Hence the lines with at least three points are exactly the lines $\rho(B)$, $B \in \mathcal{B}$, the remaining pairs span ordinary lines, and $\rho(X)$ is a configuration (not all points are collinear, as no block has 15 points) with these t_k for $k \geq 3$ and with $t_2 = |E(L)|$. Hence:

Corollary 2.5. *$m(15) \leq 8$ if and only if some PLS on 15 points with one of the six profiles of Table 1 with $t_2 \in \{7, 8\}$ and $t_8 = 0$ has an exact realization over $\mathbb{R}$.*

Note that the number of leave edges of a PLS is determined by its profile: a PLS on 15 points with profile $(t_3, t_4, \dots)$ leaves exactly $105 - \sum_{k \geq 3} \binom{k}{2} t_k$ pairs uncovered, which is 7 or 8 for the profiles in question. Moreover the degree of a point p in the leave is $\deg_L(p) = 14 - \sum_{B \ni p} (|B| - 1)$, so

$$\deg_L(p) \equiv \#\{B \in \mathcal{B} : p \in B,\ |B| \text{ even}\} \pmod 2. \tag{5}$$

(For general n the right-hand side is $n - 1$ plus the number of even blocks through p; for the controls with $n = 12, 14$ below all leave degrees are therefore odd when all blocks are odd, which is why, for instance, no leave exists for $n = 14$, $t_2 = 6$, $t_5 = 1$.)

3 Enumeration

We enumerate the PLS of Corollary 2.5 up to isomorphism (permutations of the 15 points mapping blocks to blocks) in three stages. Blocks of size at least 4 are called *big*.

Stage A: placements. For each profile we list the sets of big blocks (pairwise meeting in at most one point) up to isomorphism, by adding blocks of decreasing size one at a time and keeping one representative of each isomorphism class after each addition. Isomorphism classes are recognised with `nauty` [10] (through `pynauty`) canonical forms of the point–block incidence graph, with the block vertices coloured by block size (`placements.py`). Keeping one representative per class after each addition loses nothing, because isomorphic partial placements have isomorphic sets of extensions. The numbers of placements are given in Table 1; they are confirmed by an orbit–stabiliser count (Section 7).

Stage B: leaves. For each placement P of big blocks we enumerate, up to isomorphism of the pair (P, L), the graphs L with t_2 edges avoiding the pairs inside big blocks and satisfying the parity condition (5). The program `leaves2.c` builds L edge by edge: if some point currently has the wrong parity, it branches on the edges at the smallest such point, otherwise on all admissible

edges; candidate edges are taken modulo the automorphism group of $(P, \text{current edges})$, computed by `nauty` at every node; a partial graph with k edges is abandoned when more than $2(t_2 - k)$ points have the wrong parity; and complete graphs are reduced to a canonical form and recorded once per isomorphism class. Every isomorphism class of pairs (P, L) is produced (the standard argument: if a class is reachable through one child of a node, it is reachable through the representative of that child's orbit), and duplicates are removed by the canonical form. An independent implementation (`stageAB.py`, level-by-level generation with `pynauty` certificates) produces the same numbers for every placement of every profile considered in this paper, for $n = 12, 13, 14, 15$. The numbers of (P, L) classes are given in Table 1.

Stage C: triangle decompositions. Given (P, L), the remaining pairs of points (those neither inside a big block nor in L) must be partitioned into blocks of size 3, that is, decomposed into edge-disjoint triangles. The program `tri3.c` does this by backtracking: at each node it selects an uncovered pair $\{u, v\}$ with the fewest admissible third points w (those with $\{u, w\}$ and $\{v, w\}$ uncovered), computes with `nauty` the automorphism group of the current partial structure (big blocks, triangles chosen so far and leave edges, as a coloured incidence graph), and branches over one candidate w per orbit of that group on the candidate triangles $\{u, v, w\}$ (orbits of the group on all triples are computed, and candidates in the same orbit are identified; this is complete because every completion of the current structure covers the pair $\{u, v\}$, and an automorphism of the current structure maps a completion through $\{u, v, w\}$ to an isomorphic completion through the image triangle). Complete decompositions are reduced to canonical form (again with `nauty`) and recorded once per isomorphism class. Two different (P, L) classes cannot yield isomorphic PLS, since an isomorphism of PLS maps big blocks to big blocks and leave edges to leave edges; hence the union of the outputs over all (P, L) classes is a complete and irredundant list of the PLS with the given profile. At the leaves, and optionally at the internal nodes, `tri3.c` applies the closure tests of Section 4; in the runs reported here the tests were applied at the leaves only, so that the number of isomorphism classes rejected by each test is available (the "countall" mode).

The correctness of the search was checked in several ways (Section 7): against a brute-force variant without orbit pruning on $n = 13$, against the numbers of partial Steiner triple systems obtained in [14] with a different program, and by the positive controls of Section 7.2. The numbers of isomorphism classes per profile are given in Table 2.

4 Incidence-theorem filters

Let $\mathcal{B}$ be a PLS on X and let ρ be an exact realization over a field K. For two points $p \neq q$ of X, the line $\rho(p)\rho(q)$ contains $\rho(x)$ for the points x of the block through p, q, if there is one, and for no other point of X; in particular two distinct blocks are realized by distinct lines, and if they share a point this point is realized by the intersection of the two lines. The following consequences of the theorems of Pappus and Desargues are used as filters. We say that three distinct points $x, y, z \in X$ are *non-collinear in* $\mathcal{B}$ if they are not contained in a common block; in an exact realization such points are non-collinear.

Lemma 4.1 (Pappus filter)**.** *Let $A \neq B$ be blocks of $\mathcal{B}$, let a, b, c be distinct points of $A \setminus B$ and x, y, z distinct points of $B \setminus A$. Suppose that each of the six pairs $\{a, y\}, \{b, x\}, \{a, z\}, \{c, x\}, \{b, z\}, \{c, y\}$ lies in a block, and that the blocks through $\{a, y\}$ and $\{b, x\}$ share a point X, those*

through $\{a, z\}$ and $\{c, x\}$ share a point Y, and those through $\{b, z\}$ and $\{c, y\}$ share a point Z. If X, Y, Z are distinct and non-collinear in $\mathcal{B}$, then $\mathcal{B}$ has no exact realization over any field.

Proof. Let ρ be an exact realization; we drop ρ from the notation. The lines A and B are distinct and meet in a point o which is none of a, b, c, x, y, z (a point of X on both lines would be a common point of the two blocks, and $a, b, c \notin B$, $x, y, z \notin A$). The lines ay and bx are distinct: otherwise a, b, x, y would be collinear, the line would be A (it contains $a \neq b$), and $x \in A$. Hence $ay \cap bx$ is a single point, and it equals X because X lies on both lines. Similarly $az \cap cx = Y$ and $bz \cap cy = Z$. By Pappus's theorem, in the form "if a, b, c are distinct points of a line A and x, y, z distinct points of a line $B \neq A$, none of them equal to $A \cap B$, then the points $ay \cap bx$, $az \cap cx$, $bz \cap cy$ are collinear" (valid in $K\mathbb{P}^2$ for every field K; the script `incidence_theorems.py` verifies the underlying polynomial identity symbolically in exactly this generality), the points X, Y, Z are collinear, contradicting exactness since they are distinct and non-collinear in $\mathcal{B}$. □

Lemma 4.2 (Desargues filter)**.** *Let $o \in X$ and let ℓ_1, ℓ_2, ℓ_3 be distinct blocks through o. For $i = 1, 2, 3$ let $a_i \neq b_i$ be points of $\ell_i \setminus \{o\}$. Suppose that each of the six pairs $\{a_i, a_j\}$, $\{b_i, b_j\}$ $(i < j)$ lies in a block, and that for each $\{i, j, k\} = \{1, 2, 3\}$ the blocks through $\{a_i, a_j\}$ and $\{b_i, b_j\}$ share a point c_k. If c_1, c_2, c_3 are distinct and non-collinear in $\mathcal{B}$, then $\mathcal{B}$ has no exact realization over any field.*

Proof. In an exact realization the lines $a_i a_j$ and $b_i b_j$ are distinct: otherwise a_i, b_i, a_j, b_j would be collinear, the line would be ℓ_i (it contains $a_i \neq b_i$), and $a_j \in \ell_i \cap \ell_j = \{o\}$. So $a_i a_j \cap b_i b_j$ is a single point, equal to c_k. Write the points as vectors with $b_i = a_i + \beta_i o$, $\beta_i \neq 0$ (possible since b_i lies on the line oa_i and $b_i \neq o, a_i$). Then $\beta_j a_i - \beta_i a_j$ is a vector representing c_k (it is a combination of a_i, a_j and equals $\beta_j b_i - \beta_i b_j$), and $\beta_1 c_1 - \beta_2 c_2 + \beta_3 c_3 = 0$ for these representatives, so c_1, c_2, c_3 are collinear—this is Desargues's theorem, in a form that needs no non-degeneracy assumption on the two triangles. Again this contradicts exactness. □

In both lemmas the hypotheses are purely combinatorial and are checked exhaustively: for the Pappus filter over all pairs of blocks and all choices of ordered triples, and for the Desargues filter over all points o, all triples of blocks through o and all choices of the a_i, b_i (the program `closure_check.c`, and the same code inside `tri3.c`). A PLS that satisfies the hypotheses of one of the lemmas is called *closure-violating*; otherwise it is *closure-free*. Since the conclusion of both lemmas is a statement about a fixed set of blocks, the test can also be applied to the partial structures met during the search of Stage C: there, every added triangle is a complete block and every leave pair is final, so a triple $\{X, Y, Z\}$ is known to be non-collinear as soon as one of its pairs is a leave pair or lies in a block not containing the third point, and a violation found at a node persists in all its descendants.

5 Exact realizability

For the closure-free systems we use the algebraic test of [14], extended to blocks of arbitrary size. Let $\mathcal{B}$ be a PLS on $X = \{0, \ldots, n-1\}$ with collinear triples $\mathcal{T}$.

Construction sequence. Choose four points $e_0, e_1, e_2, e_3 \in X$ no three of which are collinear in $\mathcal{B}$ (a *frame*) and assign them the coordinates (1:0:0), (0:1:0), (0:0:1), (1:1:1). Then place the remaining points one at a time, always choosing a point p lying on the largest number of blocks that already contain two placed points:

- if p lies on two such blocks, p is the intersection of the corresponding two lines, computed by cross products (*determined* point);
- if p lies on exactly one such block, containing placed points a_1, a_2, we write $p = a_1 + t\,a_2$ with a new parameter t;
- otherwise we write $p = (1{:}t{:}t')$ with two new parameters.

Among all frames (with an early exit once at most two parameters are needed) the order with the fewest parameters m is used; every closure-free system in this paper has a frame, and $m \le 2$ for all of them.

Lemma 5.1. *If ρ is an exact realization over a field K of characteristic 0, normalised on the frame, then there is $\tau \in K^m$ such that every constructed coordinate vector, evaluated at τ, is a non-zero multiple of the corresponding point of ρ.*

Proof. Induction along the construction. For a determined point the two blocks are distinct, so the two lines are distinct in ρ (Section 4) and their cross product is a non-zero multiple of $\rho(p)$. For a point on one known block, $\rho(p) \neq \rho(a_2)$, so $\rho(p) = a_1 + ta_2$ for a unique $t \in K$. For a free point, no block contains p and two placed points; in particular $\{e_1, e_2, p\}$ is not contained in a block, so by exactness $\rho(p)$ is not on the line $x = 0$ and can be normalised to $(1{:}t{:}t')$. □

Let $I \subset \mathbb{Q}[t_1, \ldots, t_m]$ be the ideal generated by the 3×3 determinants of the coordinate vectors of all collinear triples $\mathcal{T}$ (those used in the construction vanish identically). By Lemma 5.1, an exact realization yields a point $\tau \in V(I)$ at which, in addition, the following polynomials do not all vanish: for every determined point p, its three coordinates (*well-definedness*); for every pair of points, the three 2×2 minors of their coordinate vectors (*distinctness*); and for every non-collinear triple, its determinant (*exactness*). Recall that for an ideal J, $\mathrm{sat}(I, J) = \bigcup_k (I : J^k)$ and $V(\mathrm{sat}(I, J))$ is the Zariski closure of $V(I) \setminus V(J)$; in particular $\mathrm{sat}(I, J) = (1)$ if and only if $V(I) \subseteq V(J)$ over $\overline{\mathbb{Q}}$, and then also over every field of characteristic 0. The test proceeds in three stages, implemented in Singular [15] by the script `realize2.py`:

N compute a Gröbner basis of I; if it is $\{1\}$, there is no solution at all (*no complex solution*);

D otherwise saturate successively by the well-definedness and distinctness ideals; if the result is (1), every solution is degenerate (*degenerate*);

E otherwise saturate successively by the principal ideals of the non-collinear determinants; if the result is (1), every non-degenerate solution has an extra collinearity (*extra collinearity*).

If a system survives all three stages, the remaining ideal describes the Zariski closure of the set of its exact realizations over $\mathbb{C}$ (normalised on the frame); a real point of it outside the excluded loci is an exact real realization, and its existence has to be decided separately (this happens only in the controls of Section 7.2).

Lemma 5.2. *If a PLS is closure-violating, or is rejected at stage N, D or E, then it has no exact realization over any field of characteristic 0, in particular none over $\mathbb{R}$.*

Proof. Lemmas 4.1, 4.2 and 5.1, together with the saturation property recalled above. □

6 Results

Table 2 summarises the computation for the seven profiles. Of the 23,174,731 systems, 23,056,600 are rejected by the Pappus filter, 90,690 by the Desargues filter, 6,362 at stage N and 21,078 at stage D; none is rejected at stage E, and exactly one system survives. It belongs to the profile $t_3 = 23, t_4 = 3, t_5 = 1$ and is analysed in Section 6.1: its realization space consists of two non-real conjugate points, so it has no exact realization over $\mathbb{R}$. Hence no PLS with any of the profiles has an exact real realization, and by Corollary 2.5 this proves Theorem 1.1.

profile	classes	Pappus	Desargues	free	N	D	E	S
$t_2 = 7$: $t_3 = 26,\ t_5 = 2$	55,715	55,533	173	9	5	4	0	0
$t_2 = 8$: $t_3 = 23,\ t_8 = 1$	0	0	0	0	0	0	0	0
$t_2 = 8$: $t_3 = 29,\ t_5 = 1$	1,191,890	1,150,185	26,878	14,827	3,114	11,713	0	0
$t_2 = 8$: $t_3 = 24,\ t_5 = 1,\ t_6 = 1$	2,302	2,299	3	0	0	0	0	0
$t_2 = 8$: $t_3 = 27,\ t_4 = 1,\ t_5 = 1$	3,881,944	3,838,397	34,260	9,287	2,245	7,042	0	0
$t_2 = 8$: $t_3 = 25,\ t_4 = 2,\ t_5 = 1$	7,689,223	7,664,870	21,641	2,712	769	1,943	0	0
$t_2 = 8$: $t_3 = 23,\ t_4 = 3,\ t_5 = 1$	10,353,657	10,345,316	7,735	606	229	376	0	1
total	23,174,731	23,056,600	90,690	27,441	6,362	21,078	0	1

Table 2: Isomorphism classes of partial linear spaces on 15 points with the profiles of Table 1, and their fate: rejected by the Pappus filter, by the Desargues filter (applied only to systems passing the Pappus filter), or by the algebraic stages N (no complex solution), D (degenerate), E (extra collinearity); the last column (S) counts the systems whose realization space is non-empty. The single survivor has no real point.

6.1 The surviving system

The one system that survives all tests is the PLS with blocks

$$\{2,4,5,13,14\},\quad \{2,3,6,7\},\ \{3,4,9,10\},\ \{3,5,8,11\},$$
$$\{0,1,2\},\ \{0,3,14\},\ \{0,4,8\},\ \{0,5,10\},\ \{0,6,11\},\ \{0,7,9\},\ \{0,12,13\},$$
$$\{1,3,13\},\ \{1,4,11\},\ \{1,5,9\},\ \{1,6,8\},\ \{1,7,10\},\ \{1,12,14\},$$
$$\{2,8,9\},\ \{2,10,11\},\ \{4,7,12\},\ \{5,6,12\},\ \{6,9,13\},\ \{6,10,14\},$$
$$\{7,8,14\},\ \{7,11,13\},\ \{8,10,12\},\ \{9,11,12\},$$

whose leave consists of the eight pairs $\{2,12\}$, $\{3,12\}$, $\{4,6\}$, $\{5,7\}$, $\{8,13\}$, $\{9,14\}$, $\{10,13\}$, $\{11,14\}$; the three 4-point lines are concurrent at the point 3. Its automorphism group has order 4. With the frame $0,1,3,4$ the construction sequence has a single parameter t (the point $2 = (1{:}t{:}0)$ on the line 01), and the ideal surviving stages N, D and E is generated by t^2+t+1. An independent computation with SymPy (`check_survivor.py`, which only shares the construction sequence with `realize2.py`) confirms this: the greatest common divisor of the non-trivial collinearity conditions, as polynomials in t, is $(t-1)^3(t^2+t+1)$ for the construction sequence of the data file (with another sequence, obtained from the same blocks listed in a different order, it is $t(t-1)^3(t^2+t+1)$); at the real roots $t = 1$ (and $t = 0$) several of the constructed points collapse to the zero vector, while at $t = \omega = e^{2\pi i/3}$ (and at $\bar{\omega}$) all 15 points are distinct, all collinear triples of the PLS are collinear and no other triple is. Thus the system has exactly two exact realizations normalised on the frame, conjugate under complex conjugation, and none over

$\mathbb{R}$. Scaled to Eisenstein integers, the realization at $t = \omega$ is

$0 : (1, 0, 0)$	$1 : (0, 1, 0)$	$2 : (1, \omega, 0)$
$3 : (0, 0, 1)$	$4 : (1, 1, 1)$	$5 : (\omega, 1, 1 + \omega)$
$6 : (1 + \omega, -1, -\omega)$	$7 : (-2 - \omega, 1 - \omega, -1 - 2\omega)$	$8 : (-2 - \omega, 1 + 2\omega, 1 + 2\omega)$
$9 : (-1, -1, \omega)$	$10 : (-1 - \omega, -1 - \omega, -\omega)$	$11 : (-2 - \omega, 1 + 2\omega, -2 - \omega)$
$12 : (1 - \omega, -1 - 2\omega, -\omega)$	$13 : (0, -1 + \omega, -1)$	$14 : (-1 + \omega, 0, \omega)$

(`verify_realization.py` checks exactness in $\mathbb{Q}(\omega)$). There is also a conceptual reason for the absence of real realizations. The eight points $0, 2, 3, 4, 7, 8, 9, 14$ carry exactly eight collinear triples of the system, namely $\{0, 3, 14\}$, $\{0, 4, 8\}$, $\{0, 7, 9\}$, $\{2, 3, 7\}$, $\{2, 4, 14\}$, $\{2, 8, 9\}$, $\{3, 4, 9\}$, $\{7, 8, 14\}$, three through each point: they form a Möbius–Kantor configuration (8_3) (the other three copies are $\{0, 2, 3, 5, 6, 10, 11, 14\}$, $\{1, 2, 3, 4, 7, 10, 11, 13\}$ and $\{1, 2, 3, 5, 6, 8, 9, 13\}$). The (8_3) configuration is unique as an incidence structure, and an exact realization of it (eight distinct points with exactly these eight collinear triples) over a field K of characteristic 0 exists if and only if $\sqrt{-3} \in K$ (see e.g. [8, Section 2.2]; the script `verification/mk_check.py` recomputes the realization space $s^2 - s + 1 = 0$ of (8_3) and checks that the four 8-point subsystems are isomorphic to it). An exact real realization of the system would restrict to an exact real realization of (8_3) on these eight points. This proves Theorem 1.2.

The Böröczky example shows that the bound is sharp. Explicitly, let X_{16} consist of the vertices $(\cos(2\pi k/8), \sin(2\pi k/8))$ of a regular octagon, $k = 0, \dots, 7$, and the points at infinity in the directions $(\cos(\pi k/8), \sin(\pi k/8))$, $k = 0, \dots, 7$. The script `boroczky.py` verifies in exact arithmetic (in $\mathbb{Q}(\sqrt{2})$) that X_{16} has line profile $t_2 = 8$, $t_3 = 28$, $t_8 = 1$, and that deleting the point at infinity in the direction $(1, 0)$ (the direction of a diagonal of the octagon with vertex $(1, 0)$) leaves a 15-point set with profile $t_2 = 9$, $t_3 = 25$, $t_7 = 1$. (Deleting a point at infinity in the direction of a side gives 12 ordinary lines instead.) Together with Theorem 1.1, $m(15) = 9$.

7 Verification

7.1 Consistency checks

Every stage of the computation was checked in at least two independent ways. The scripts are in the directory `verification/` of the accompanying material; several of them were written independently from the statements in this paper, without consulting the original code, and only afterwards compared with the programs of Sections 3–5.

Stage A. For each profile, the number of labelled placements (sets of big blocks with the prescribed sizes, pairwise meeting in at most one point, on the labelled point set) was counted by exhaustive enumeration (`count_placements.c`, `count_placements2.c`) and compared with $\sum_P 15!/|\operatorname{Aut}(P)|$ over the placement classes of Table 1, with $|\operatorname{Aut}(P)|$ from `nauty`: the two numbers agree for all seven profiles (`os_stageA.py`; for instance 78,835,056,300 labelled placements for $t_4 = 3, t_5 = 1$). By the orbit–stabiliser theorem this identity holds if and only if the list of classes is complete and irredundant.

Stage B. The two implementations `stageAB.py` and `leaves2.c` use different generation strategies (level-by-level certificate deduplication versus orbit pruning with leaf canonical forms) and different canonical-form back ends (`pynauty` certificates versus `nauty` canonical labellings);

they produce the same numbers of (P, L) classes for every placement of every profile treated in this paper, for $n = 12, 13, 14, 15$. In addition, for every placement P of every profile for $n = 15$ the number of labelled leaves (computed by a character sum over the parity constraints) equals $\sum_L |\operatorname{Aut}(P)|/|\operatorname{Aut}(P, L)|$ over the (P, L) classes (`osb.py`), so the 252,231 classes are complete and irredundant.

Stage C. (i) The variant `tri_ns.c` without orbit pruning (all candidates tried, duplicates removed only by canonical forms at the leaves) produces exactly the same sets of canonical forms as the pruned search on all profiles for $n = 13$. (ii) An independent brute-force implementation in Python (`brute_stageC.py`: all labelled triangle decompositions, no pruning, `pynauty` certificates) produces the same set of isomorphism classes as `tri3.c` on 491 (P, L) lines for $n = 12$ and $n = 13$ (all lines whose brute-force enumeration finishes within five minutes). (iii) For every (P, L) line of the profiles $t_5 = 2$ and $t_5 = 1, t_6 = 1$ for $n = 15$, and for 48 of the 60 lines of the profile $t_3 = 29, t_5 = 1$ (those whose labelled decompositions could be counted within two minutes), the number of labelled triangle decompositions of the uncovered pairs, counted by an exhaustive search without any symmetry reduction (`count_dec.c`), equals $\sum |\operatorname{Aut}(P, L)|/|\operatorname{Aut}(\mathcal{B})|$ over the classes $\mathcal{B}$ found by `tri3.c` (`os_stageC2.py`); this covers lines with $|\operatorname{Aut}(P, L)|$ up to 1,935,360 and up to 81,768,960 labelled decompositions. (iv) The class counts and closure verdicts of the three profiles that were first enumerated with the earlier program `tri.c` (all classes written out) and filtered with `closure_check.c` were reproduced by `tri3.c` in "countall" mode, with identical sets of closure-free classes. (v) For the all-triples profiles the search reproduces the numbers of partial Steiner triple systems with prescribed leave obtained in [14] with an independent program (the 267 systems PSTS(13, 24) with an even 6-edge leave and the 787 systems PSTS(14, 28) with a perfect-matching leave).

Closure filters. (i) A witness-printing variant of the filter (`closure_witness.c`) outputs, for every rejected system, the blocks and points of the Pappus or Desargues instance it found, and `verify_witness.py`, written from the statements of Lemmas 4.1 and 4.2, checks every hypothesis of the lemma for that witness: 136,709 Pappus witnesses and 1,931 Desargues witnesses (all systems of the profile $t_5 = 1, t_6 = 1$, every 20th class of $t_3 = 29, t_5 = 1$ and every 50th class of $t_3 = 27, t_4 = 1, t_5 = 1$) were verified in this way, and all 27,441 closure-free systems were confirmed to contain no instance. (ii) An independent exhaustive Python implementation of the two lemmas (`indep_closure.py`) agrees with the C verdicts on 8,250 classes. (iii) The C implementation and the original Python implementation (`realize2.py`) agree class by class on all 2,994 PLS classes for $n = 13$, and the survivors written by `tri3.c` coincide with the survivors of the Python filter for every profile for $n = 13$ and for the first three (P, L) classes of the profile $t_3 = 29, t_5 = 1$ in all modes of the program (tests at the leaves only, incremental or full tests at internal nodes). (iv) The filters accept the collinearity structures of 30 exact real configurations with 12 to 16 points (Böröczky sets and subsets, subsets of the 4×4 grid with points at infinity, points on a cubic, `realconfigs.py`), and the algebraic test returns a non-empty realization space for each of them. (v) The algebraic test was run without the closure filters on all 2,994 classes for $n = 13$: every closure-violating class is rejected at stage N or D (1,022 and 1,971 classes in all), and the verdicts on the closure-free classes are unchanged. For $n = 15$ the same was done for a random sample of 1,000 closure-violating classes of the profile $t_3 = 29, t_5 = 1$ (600 rejected by Pappus and 400 by Desargues): all are rejected at stage N or D (168 and 832 respectively).

Algebraic test. (i) Every rejection is a certificate that a certain ideal is the unit ideal, so an error could only produce a false rejection through a software error in SINGULAR or in the generated script; the scripts are generated by a small, fully deterministic writer. (ii) An independently written realizability script (`indep_alg.py`: last admissible frame instead of the first, a different construction order without parameter minimisation, a different normalisation of free points, one saturation chain over all non-degeneracy ideals) was run on the closure-free systems: 27,439 of the 27,441 systems were rejected by it as well, the survivor of Section 6.1 was confirmed, and 1 system exceeded the time limit allowed for this check. (iii) Independently of any Gröbner-basis computation, a configuration of points in $\mathbb{RP}^2$ induces a rank-3 chirotope whose zero set is exactly the set of collinear triples; the existence of such a chirotope (equivalently, by the Folkman–Lawrence theorem, of an arrangement of pseudolines with the prescribed incidences) is decided by a SAT solver (`omcnf2.c`, using the three-term Grassmann–Plücker relations and CaDiCaL), a method that shares no code and no mathematics with the algebraic test. Applied to all 27,441 closure-free systems, the solver reports that none of them admits a chirotope, i.e. none is realizable even by pseudolines—including the survivor of Section 6.1, as one expects from its Möbius–Kantor subconfigurations. Thus every algebraic rejection is independently confirmed, and the proof of Theorem 1.1 can be read with the SAT solver in place of the Gröbner-basis computations. (iv) The survivors (Section 6.1 and the controls below) are analysed with SymPy, independently of SINGULAR.

7.2 Positive controls

The same pipeline, run with $n = 12$, 13, 14 points and $t_2 = m(n)$, must find the known extremal configurations, and run with $t_2 = m(n) - 1$ it must find nothing. Table 3 gives the numbers. For $n = 13$ the reduction gives five profiles ($t_3 = 24$; $t_3 = 22, t_4 = 1$; $t_3 = 20, t_4 = 2$; $t_3 = 18, t_4 = 3$; $t_3 = 19, t_6 = 1$), 2,994 PLS classes in all, of which exactly one survives all tests: it has profile $t_3 = 18, t_4 = 3$, its realization space (normalised on a frame) is the zero set of $t^2 - t - 1$, and both roots give exact real realizations. This is McKee's configuration [2], whose coordinates indeed involve the golden ratio, and we obtain as a by-product that it is the only 13-point configuration with 6 ordinary lines up to projective equivalence and the Galois conjugation exchanging the two roots. For $n = 12$ the reduction gives five profiles ($t_3 = 20$; $t_3 = 18, t_4 = 1$; $t_3 = 16, t_4 = 2$; $t_3 = 14, t_4 = 3$; $t_3 = 15, t_6 = 1$), 320 PLS classes in all, and exactly two survive: one with profile $t_3 = 15, t_6 = 1$ and a unique, rational realization ($t = 1/2$), which is projectively equivalent to Böröczky's X_{12} (a regular hexagon with the six points at infinity of its sides and diagonals), and one with profile $t_3 = 18, t_4 = 1$ and realization space $t^2 - t - 1 = 0$, which is McKee's configuration with one point removed. The script `verify_realization.py` rebuilds the three realizations along the construction sequence in exact arithmetic (in $\mathbb{Q}$ or $\mathbb{Q}(\sqrt{5})$) and checks that all points are distinct, that all collinear triples of the PLS are collinear and that no other triple is collinear. Thus the pipeline finds exactly the known configurations with 6 ordinary lines for $n = 12$ and $n = 13$, and in both cases it shows that there are no others.

For $n = 14$ and $t_2 = 6$ (one less than $m(14) = 7$) the reduction gives two profiles; one of them admits no leave at all, and the 1,139 systems of the other are all rejected. Note that the profile lemma for $n = 14$ only uses the Kelly–Moser bound $t_2 \geq 6$, so this is an independent confirmation of $m(14) \geq 7$, which originally required the Csima–Sawyer theorem. For $n = 14$ and $t_2 = 7$ there are eight profiles (including $t_3 = 28$, whose 787 systems are exactly the partial Steiner triple systems PSTS$(14, 28)$ with a perfect-matching leave found in [14]) and 498,803 systems, of which exactly one survives: it has profile $t_3 = 21$, $t_7 = 1$, its realization space is

$\{t_0 = t_1 + 1,\ t_1^3 + 2t_1^2 - t_1 - 1 = 0\}$, and the three roots $t_1 = -1 - 2\cos(2\pi k/7)$, $k = 1, 2, 3$, are real and give exact real realizations—this is Böröczky's X_{14} (a regular heptagon with the seven points at infinity of its sides and diagonals) and its two Galois conjugates, in which the heptagon is replaced by the star heptagons. So X_{14} is the only 14-point configuration with 7 ordinary lines, up to projective equivalence and Galois conjugation.

n	t_2	profile	(P, L)	classes	Pappus	Desargues	free	N	D	E	S
12	6	$t_3 = 20$	1	5	4	0	1	0	1	0	0
12	6	$t_3 = 18,\ t_4 = 1$	7	65	45	8	12	3	8	0	1
12	6	$t_3 = 16,\ t_4 = 2$	52	115	92	3	20	13	7	0	0
12	6	$t_3 = 14,\ t_4 = 3$	240	129	106	5	18	13	5	0	0
12	6	$t_3 = 15,\ t_6 = 1$	11	6	2	0	4	2	1	0	1
13	6	$t_3 = 24$	3	267	238	3	26	6	20	0	0
13	6	$t_3 = 22,\ t_4 = 1$	5	590	538	19	33	9	24	0	0
13	6	$t_3 = 20,\ t_4 = 2$	30	987	922	33	32	11	21	0	0
13	6	$t_3 = 18,\ t_4 = 3$	145	1,148	1,084	33	31	18	12	0	1
13	6	$t_3 = 19,\ t_6 = 1$	3	2	2	0	0	0	0	0	0
14	6	$t_3 = 25,\ t_5 = 1$	0	0	0	0	0	0	0	0	0
14	6	$t_3 = 23,\ t_4 = 1,\ t_5 = 1$	12	1,139	1,124	9	6	3	3	0	0
14	7	$t_3 = 28$	1	787	345	253	189	8	180	1	0
14	7	$t_3 = 26,\ t_4 = 1$	7	25,662	23,398	1,119	1,145	216	929	0	0
14	7	$t_3 = 24,\ t_4 = 2$	126	119,196	115,653	2,106	1,437	501	936	0	0
14	7	$t_3 = 22,\ t_4 = 3$	1,610	190,276	187,766	1,800	710	272	438	0	0
14	7	$t_3 = 20,\ t_4 = 4$	13,796	161,846	160,924	776	146	46	100	0	0
14	7	$t_3 = 23,\ t_6 = 1$	19	226	153	65	8	5	3	0	0
14	7	$t_3 = 21,\ t_4 = 1,\ t_6 = 1$	294	803	792	10	1	0	1	0	0
14	7	$t_3 = 21,\ t_7 = 1$	1	7	0	4	3	0	2	0	1

Table 3: The controls: all profiles for $n = 12, 13, 14$ and $t_2 = m(n)$ (and $t_2 = m(n)-1$ for $n = 14$), with the same columns as Table 2. The four surviving systems are X_{12}, McKee's configuration minus a point, McKee's configuration and X_{14}; all four have real realizations.

8 Discussion

Theorem 1.1 determines the first unknown term of the sequence $m(n)$. The next open cases are $m(17) \in \{8, \dots, 12\}$ and $m(19) \in \{9, \dots, 12\}$, where (1) predicts 12 in both cases; for $n = 17$ the gap is large, and the number of profiles and the size of the enumeration grow quickly with n and with t_2, so new ideas (for instance a search organised around the dual arrangement, or stronger combinatorial necessary conditions) seem necessary to go much further with the present approach. A different natural continuation is the classification of all 15-point configurations with exactly 9 ordinary lines, which requires the profiles with $t_2 = 9$ (among them $t_3 = 32$, excluded by [14]); at least two such configurations exist, Böröczky's and the dual of the simplicial arrangement $A(15, 5)$.

Our filters use the theorems of Pappus and Desargues and are therefore not valid for arrangements of pseudolines. The Kelly–Moser and Csima–Sawyer bounds, being consequences of Euler's formula and local counting, do hold for pseudoline arrangements, and it is natural to ask whether the pseudoline analogue $m^{ps}(15)$ (the minimum number of double points of an arrangement of 15 pseudolines) is also 9. The SAT computation of Section 7.1 shows that none

of the 27,441 closure-free systems is realizable by pseudolines, but a closure-violating PLS may well be, and deciding this for all 23,174,731 systems would take an estimated 500 CPU hours with the present encoding.

9 Programs and data

All programs and data are provided with this paper. The main components are: `profiles.py` (Lemma 2.2); `placements.py` and `leaves2.c` (Stages A, B), with `stageAB.py` as an independent implementation; `tri3.c` (Stage C with closure tests), with `tri.c` and `tri_ns.c` as reference implementations; `closure_check.c`; `realize2.py` (the algebraic test, generating SINGULAR scripts); `analyze_sol.py` and `verify_realization.py` (survivors); `incidence_theorems.py`; `boroczky.py`; `pipeline.sh` (the complete pipeline for given n and t_2). The data comprise, for every profile, the lists of (P, L) classes, the closure-free PLS classes and their verdicts, and the per-line logs of the search with node, leaf and class counts; the complete lists of PLS classes for the two largest profiles are not included (about 5 GB of text) but are regenerated by `tri3.c` in a few hours on one core. Software used: `nauty` 2.8 [10], `pynauty`, SINGULAR 4.3 [15], SymPy. The whole computation for $n = 15$ takes about five CPU hours on one core of a laptop-class machine.

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